



MediNova Nexus

P V T . L T D .

— Smart Care, Connected Life —

SMART HOSPITAL MANAGEMENT SYSTEM

PROJECT REPORT

SUBMITTED TO:

Department of Computer Science & Engineering
Centurion University of Technology and Management

SUBMITTED BY:

Jagannath Mohapatra
Reg. No: 230301120151
B.Tech (CSE)
Mail ID: omminutescraft@gmail.com
Mobile No: 9602711070

COMPANY NAME:

MediNova Nexus Pvt. Ltd.

GUIDED BY:

Dr NIBEDITA ADHIKARI

DATE:

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SOFTWARE DESIGN DOCUMENT (SDD)

Project: Smart Hospital Management System

Company: Mediniva Nexus Pvt Ltd

1. INTRODUCTION

1.1 Purpose

This Software Design Document (SDD) provides a comprehensive and detailed design of the Smart Hospital Management System developed by Mediniva Nexus Pvt Ltd. It describes system architecture, components, database structure, interfaces, and internal working of modules.

1.2 Scope

The system is designed to:

- Manage hospital operations digitally**
- Provide patient, doctor, pharmacy, and admin services**

- Enable appointment booking, diagnosis, billing, and delivery
- Ensure secure and efficient healthcare data management

2. SYSTEM OVERVIEW

2.1 System Description

The system is a web-based centralized platform that connects:

- Patients
- Doctors
- Admin
- Pharmacy
- Laboratory
- Delivery system

2.2 System Architecture

Architecture Type: Three-Tier Architecture

Layers:

- 1. Presentation Layer (Frontend)**
- 2. Application Layer (Backend)**
- 3. Data Layer (Database)**

3. ARCHITECTURAL DESIGN

3.1 High-Level Architecture

- Client (Browser / App)**
- API Server**
- Database Server**

Flow:

User → Frontend → Backend API → Database → Response

3.2 Component Diagram Description

Components:

- **User Interface Module**
- **Authentication Module**
- **Appointment Module**
- **Medical Records Module**
- **Billing System**
- **Pharmacy Module**
- **Delivery System**
- **Notification System**

4. MODULE DESIGN

4.1 Patient Module

Features:

- **Registration/Login**
- **Profile Management**
- **Book Appointment**

- **View Prescription**
- **Order Medicines**
- **Online Payment**

Inputs:

- **Name, Age, Gender**
- **Symptoms**
- **Appointment date**

Outputs:

- **Appointment confirmation**
- **Medical history**
- **Bills**

4.2 Doctor Module

Features:

- **Login**

- **View Appointments**
- **Diagnose Patients**
- **Write Prescription**
- **Upload Reports**

Inputs:

- **Patient details**
- **Diagnosis notes**

Outputs:

- **Prescription**
- **Medical advice**

4.3 Admin Module

Features:

- **Manage Users**
- **Manage Doctors**

- **View Reports**
- **System Monitoring**

Responsibilities:

- **Data control**
- **System maintenance**
- **Security control**

4.4 Pharmacy Module

Features:

- **Medicine inventory**
- **Order processing**
- **Stock updates**

4.5 Laboratory Module

Features:

- Test booking
- Report generation
- Upload results

4.6 Delivery Module

Features:

- Assign delivery agent
- Track delivery
- Update delivery status

5. DATABASE DESIGN

5.1 Database Type

Relational Database (MySQL / PostgreSQL)

5.2 Tables Structure

1. Patient Table

- **Patient_ID (PK)**
- **Name**
- **Age**
- **Gender**
- **Contact**
- **Address**

2. Doctor Table

- **Doctor_ID (PK)**
- **Name**
- **Specialization**
- **Experience**
- **Contact**

3. Appointment Table

- **Appointment_ID (PK)**
- **Patient_ID (FK)**
- **Doctor_ID (FK)**
- **Date**
- **Status**

4. Prescription Table

- **Prescription_ID**
- **Doctor_ID**
- **Patient_ID**
- **Medicines**
- **Notes**

5. Billing Table

- **Bill_ID**
- **Patient_ID**
- **Amount**
- **Payment_Status**

6. Medicine Table

- **Medicine_ID**
- **Name**
- **Price**
- **Stock**

5.3 ER Relationship

- **Patient ↔ Appointment (1:M)**
- **Doctor ↔ Appointment (1:M)**
- **Patient ↔ Prescription (1:M)**

6. INTERFACE DESIGN

6.1 User Interface

Patient UI:

- **Dashboard**

- **Book Appointment**
- **Medical Records**

Doctor UI:

- **Appointment List**
- **Prescription Panel**

Admin UI:

- **Analytics Dashboard**
- **Control Panel**

6.2 API Design

Example APIs:

- **POST /login**
- **GET /appointments**
- **POST /book-appointment**
- **GET /prescription**

7. DATA FLOW DESIGN

Level 0:

User → System → Database → Output

Level 1:

- Patient → Appointment → Doctor**
- Doctor → Prescription → Patient**
- Patient → Payment → Billing**

8. SECURITY DESIGN

8.1 Authentication

- Username & Password**
- OTP verification**

8.2 Authorization

- Role-based access**
- Admin full control**

8.3 Data Security

- Encryption (HTTPS)**
- Secure database storage**

9. ERROR HANDLING

- Invalid login → Error message**
- Payment failure → Retry option**
- Server error → Notification**

10. PERFORMANCE DESIGN

- Fast response time (< 2 seconds)
- Load handling (multiple users)
- Optimized queries
